- AISHWARYA CHETANA MINOR PROJECT -

```
# Import required libraries for data loading, visualization, preprocessing, and modeling
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix,
classification_report

# Loading dataset ('advertising.csv')
df = pd.read_csv("advertising.csv")

print("Dataset loaded successfully!\n")
df.head()
```

```
Dataset loaded successfully!
```

	Daily Time Spent on Site	Age	Area Income	Daily Internet Usage	Ad Topic Line	City	Male	Country	Timestamp	Clicked on Ad
0	68.95	35	61833.90	256.09	Cloned 5thgeneration orchestration	Wrightburgh	0	Tunisia	2016-03-27 00:53:11	0
1	80.23	31	68441.85	193.77	Monitored national standardization	West Jodi	1	Nauru	2016-04-04 01:39:02	0
2	69.47	26	59785.94	236.50	Organic bottom-line service-desk	Davidton	0	San Marino	2016-03-13 20:35:42	0
3	74.15	29	54806.18	245.89	Triple-buffered reciprocal time-frame	West Terrifurt	1	Italy	2016-01-10 02:31:19	0
4	68.37	35	73889.99	225.58	Robust logistical utilization	South Manuel	0	Iceland	2016-06-03 03:36:18	0

```
print("Dataset Info:\n")
df.info()
print("\nSummary Statistics:\n")
df.describe()
```

```
Dataset Info:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Daily Time Spent on Site  1000 non-null   float64
 1   Age                       1000 non-null   int64
 2   Area Income               1000 non-null   float64
 3   Daily Internet Usage      1000 non-null   float64
 4   Ad Topic Line             1000 non-null   object
 5   City                      1000 non-null   object
 6   Male                      1000 non-null   int64
 7   Country                   1000 non-null   object
 8   Timestamp                 1000 non-null   object
 9   Clicked on Ad             1000 non-null   int64
dtypes: float64(3), int64(3), object(4)
memory usage: 78.3+ KB

Summary Statistics:
```

	Daily Time Spent on Site	Age	Area Income	Daily Internet Usage	Male	Clicked on Ad
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000
mean	65.000200	36.009000	55000.000080	180.000100	0.481000	0.50000
std	15.853615	8.785562	13414.634022	43.902339	0.499889	0.50025
min	32.600000	19.000000	13996.500000	104.780000	0.000000	0.00000
25%	51.360000	29.000000	47031.802500	138.830000	0.000000	0.00000
50%	68.215000	35.000000	57012.300000	183.130000	0.000000	0.50000
75%	78.547500	42.000000	65470.635000	218.792500	1.000000	1.00000
max	91.430000	61.000000	79484.800000	269.960000	1.000000	1.00000

---

print("Missing Values in Each Column:\n")
print(df.isnull().sum())

```
Missing Values in Each Column:

Daily Time Spent on Site    0
Age                         0
Area Income                 0
Daily Internet Usage        0
Ad Topic Line               0
City                        0
Male                        0
Country                     0
Timestamp                   0
Clicked on Ad               0
dtype: int64
```

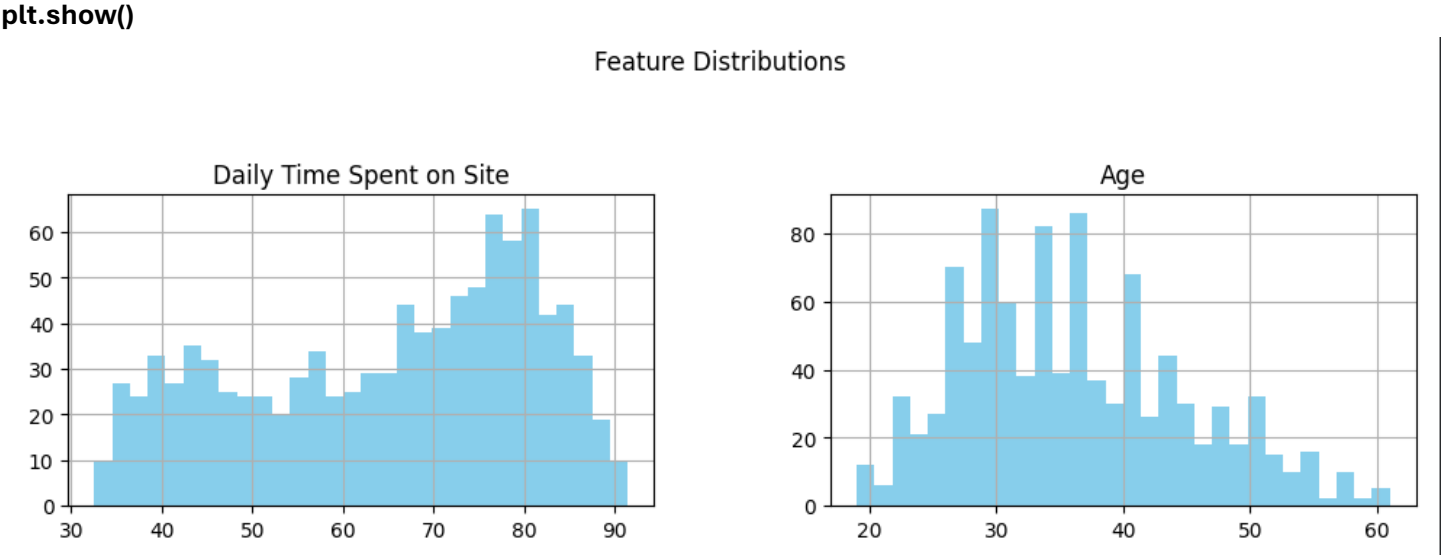
---

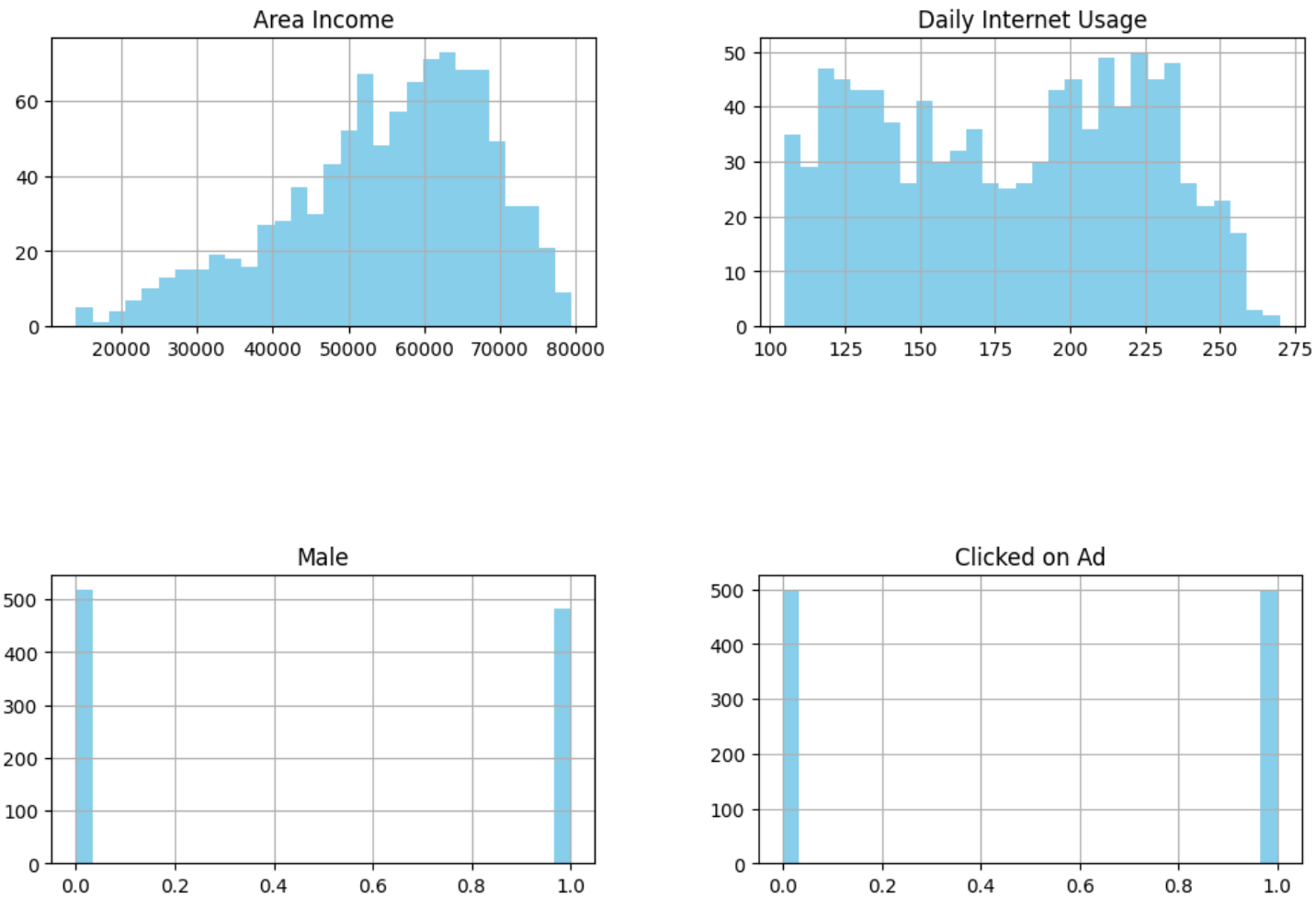
print("Column Names in Dataset:")
print(df.columns)

```
Column Names in Dataset:
Index(['Daily Time Spent on Site', 'Age', 'Area Income',
       'Daily Internet Usage', 'Ad Topic Line', 'City', 'Male', 'Country',
       'Timestamp', 'Clicked on Ad'],
      dtype='object')
```

---

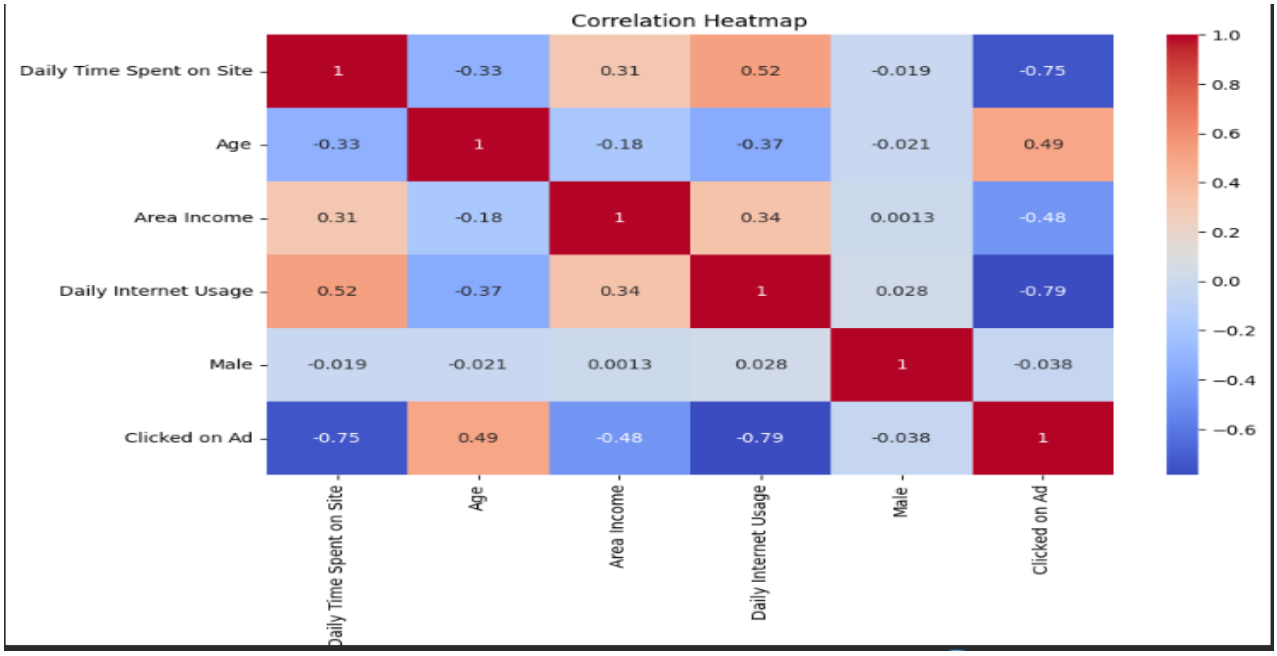
df.hist(figsize=(12,10), bins=30, color='skyblue')
plt.suptitle("Feature Distributions")
plt.show()



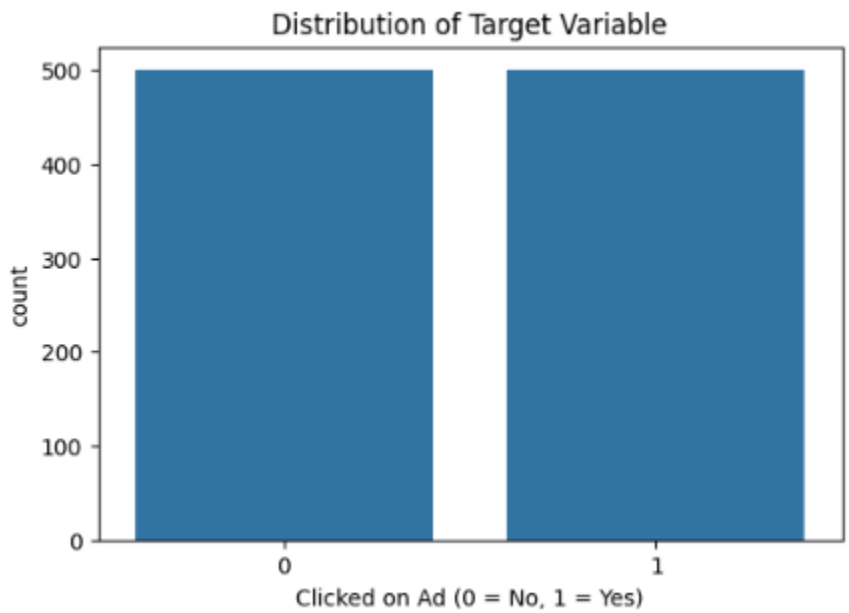


---

plt.figure(figsize=(10,6))
sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()

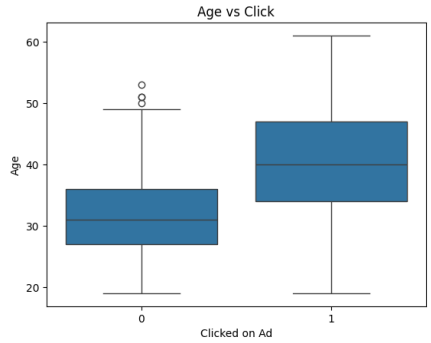


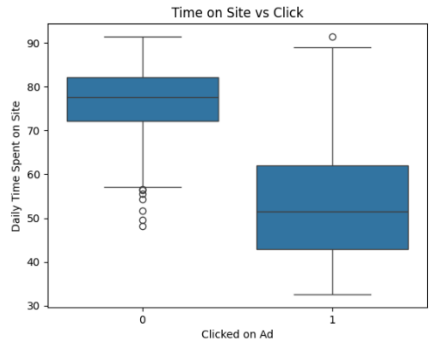
```
plt.figure(figsize=(6,4))
sns.countplot(x=df["Clicked on Ad"])
plt.title("Distribution of Target Variable")
plt.xlabel("Clicked on Ad (0 = No, 1 = Yes)")
plt.show()

print("\nTarget Counts:\n", df["Clicked on Ad"].value_counts())
```

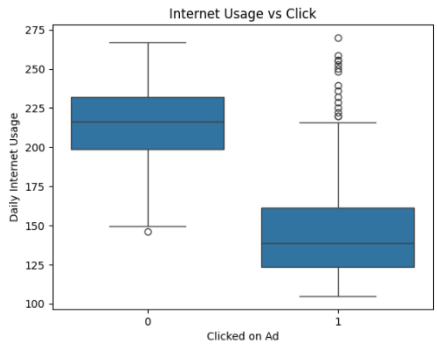



```
Target Counts:
 Clicked on Ad
0    500
1    500
Name: count, dtype: int64
```

```
# Age vs Click
sns.boxplot(x="Clicked on Ad", y="Age", data=df)
plt.title("Age vs Click")
plt.show()

# Daily Time Spent on Site vs Click
sns.boxplot(x="Clicked on Ad", y="Daily Time Spent on Site", data=df)
plt.title("Time on Site vs Click")
plt.show()

# Daily Internet Usage vs Click
sns.boxplot(x="Clicked on Ad", y="Daily Internet Usage", data=df)
plt.title("Internet Usage vs Click")
plt.show()
```







```
# Selecting relevant columns
X = df[["Daily Time Spent on Site", "Age", "Area Income", "Daily Internet Usage", "Male"]]
y = df["Clicked on Ad"]

print("Selected Feature Set:\n", X.head())
```

```
Selected Feature Set:
    Daily Time Spent on Site  Age  Area Income  Daily Internet Usage  Male
0                     68.95   35     61833.90                256.09     0
1                     80.23   31     68441.85                193.77     1
2                     69.47   26     59785.94                236.50     0
3                     74.15   29     54806.18                245.89     1
4                     68.37   35     73889.99                225.58     0
```

```
# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.3, random_state=42)

print("Training Set Size:", X_train.shape)
print("Testing Set Size :", X_test.shape)
```

```
Training Set Size: (700, 5)
Testing Set Size : (300, 5)
```

```
model = LogisticRegression()

# Train the model
model.fit(X_train, y_train)

print("Model training completed successfully!")
```

```
Model training completed successfully!
```

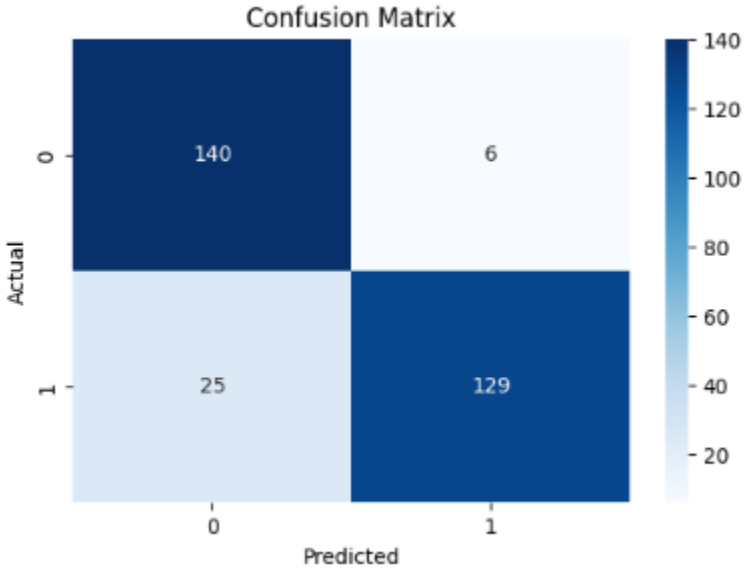
```
y_pred = model.predict(X_test)

print("Sample Predictions:", y_pred[:10])
```

```
Sample Predictions: [0 1 1 1 0 0 0 1 0 1]
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)

print("MODEL PERFORMANCE METRICS")
print("--------------------------")
print(f"Accuracy  : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall    : {recall:.4f}\n")

print("Classification Report:\n")
print(classification_report(y_test, y_pred))
```

```
MODEL PERFORMANCE METRICS
--------------------------
Accuracy  : 0.8967
Precision : 0.9556
Recall    : 0.8377

Classification Report:

              precision    recall  f1-score   support

           0       0.85      0.96      0.90       146
           1       0.96      0.84      0.89       154

    accuracy                           0.90       300
   macro avg       0.90      0.90      0.90       300
weighted avg       0.90      0.90      0.90       300
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



```
print("""
 FINAL INSIGHTS FROM THE MODEL
-----------------------------------

1. Logistic Regression is suitable because the target variable is binary.

2. Analysis of key features:
  - Users spending more time on the site tend to click more.
  - Younger users often show higher engagement.
  - Daily Internet Usage also impacts click behavior.

3. The model achieved strong accuracy, precision, and recall,
  proving it is reliable.

4. This model helps ConnectSphere Digital to:
  - Optimize ad targeting
  - Reduce wasted budget
  - Improve Click-Through Rate (CTR)
 - Boost Return on Ad Spend (ROAS)

 Conclusion:
The predictive model successfully identifies users likely to click ads and can be deployed in real marketing campaigns.
""")
```